

Better executives work at better firms: A network method for measuring sorting

Miklós Koren^{1,2,4,5*}, Ulrich Wohak¹, Krisztina Orbán³

¹Central European University, Vienna, Austria.

²ELTE Centre for Economic and Regional Studies, Budapest, Hungary.

³Monash University, Melbourne, Australia.

⁴Centre for Economic Policy Research, London, UK.

⁵CESifo, Munich, Germany.

*Corresponding author. Email: miklos@koren.work

Markets may allocate better executives to better firms, but sparse executive mobility makes this sorting difficult to measure. Our match-level leave-out comparison retains only 4.6% of executives after its mobility restrictions. We instead model firm and executive effects as a Gaussian Markov random field on the mobility network. The decline in outcome covariance with network distance identifies sorting. Applied to nearly the full Hungarian executive network, the estimator yields a positive correlation, $\hat{\rho} = 0.32$; by contrast, fixed effects on their admissible subgraph yield a correlation of -0.59 . Firm effects account for 47.7% of revenue variance, executive effects for 3.8%, and their sorting covariance for 8.9%. Random executive assignment would lower aggregate revenue by 16.5%.

The allocation of resources across firms can affect aggregate economic productivity (1). In this paper we ask whether business executives, who are arguably an important resource for firm performance (2), are allocated to firms in a way that maximizes their contribution. Several executive-pay studies assume perfect sorting of good executives to good firms and interprets compensation dispersion as the competitive price of scarce talent (3, 4). There is, however, little direct evidence on the correlation between firm and executive quality.

Motivated by two-way fixed-effects methods on matched employer–employee data (5, 6), empirical management studies estimate “CEO effects” on performance in an empirical design based on CEO mobility across firms (7–10). These methods work best on dense mobility networks, but executive careers are short and their network is fragmented: three quarters of executives in our data never move to a second firm. Limited-mobility bias is therefore severe, and existing corrections require connected, densely linked samples (11–13).¹

We develop a random-effects estimator for sparse mobility networks like that of executives. Instead of estimating millions of firm and executive effects, we estimate four parameters of their joint distribution: firm heterogeneity, executive heterogeneity, sorting, and match-specific noise. A Gaussian Markov random field links these latent effects through the observed mobility network (17).

The key identification challenge is to separate sorting from the scale of firm and executive heterogeneity. We use a full-information maximum likelihood estimator, but the intuition for identification is based on how covariance between firm outcomes decays along a network path. Two jobs can have correlated outcomes even when they share neither a firm nor an executive, because correlation propagates along the path connecting them. If executives are sorted to firms, then two executives who worked at the same firm will be similar (because they are both similar to the firm). And two executives who are linked by someone else, will also be indirectly correlated, albeit more weakly. The correlation declines with the number of edges separating them, and the decay rate identifies the magnitude of sorting.

Unlike existing fixed-effects methods, which can only be estimated on a connected component, our estimator uses data from the full graph, including disconnected components and isolated firm–executive pairs. This increased coverage does not only increase statistical power, but also provides a more representative estimate of sorting across the economy. As we show below, the largest connected component provides a biased sample of firms and executives.

Using data on the universe of corporations and their executives in Hungary, we find positive sorting of executives to firms, $\hat{\rho} = 0.32$. This is in contrast to other estimators and other methods that report much weaker, or even negative correlation between worker and firm quality (13). Firm effects account for 47.7% of revenue variance, executive effects for 3.8%, and the covariance for

¹When the outcome is wages, the estimated covariance can combine production complementarities with rent sharing (14–16). We study firm revenue, so this particular confounding does not arise.

8.9%. Random executive assignment would lower aggregate revenue by 16.5%. The sorting estimate is stable across various specifications for pruning the mobility network, or splitting the data into subsamples.

We validate the estimator on a dense worker-firm dataset, the Veneto Employer History, where it reproduces the leave-one-out estimate of (13). Leave-one-out methods are ill suited for sparse networks, partly because they discard much of the data, but also because the weighting of the remaining observations increases estimation error (18). On our sample, simple plug-in fixed effects give $\hat{\rho} = -0.59$, the leave-one-out correction gives -0.16 , against our estimate of $+0.32$.

A network model of sorting

We start from a multiplicative model of firm performance, where firm productivity and executive skill are complementary inputs (2). Let Y_{im} denote the revenue of firm i when it is managed by executive m , A_i firm productivity (or other persistent performance shifter), and Z_m executive skill. Allowing for an uncorrelated total-factor-productivity shock ε_{im} , we have

$$Y_{im} = A_i Z_m \exp(\varepsilon_{im}).$$

The positive cross derivative of the production function with respect to firm and executive quality implies that the optimal allocation requires positive sorting: better executives should be matched to better firms. If that is the case, observed matches will have positively correlated latent effects.

Taking logs, we can decompose the firm outcome into a persistent firm and executive effect plus a match-specific disturbance:

$$y_{im} = a_i + z_m + \varepsilon_{im}, \tag{1}$$

Fixed-effects methods estimate every a_i and z_m separately. We instead model their distribution as jointly normal.

Let $\mathbf{x} = (a_1, \dots, a_{N_f}, z_1, \dots, z_{N_m})^\top$ collect the firm and executive effects. We model these effects as a Gaussian Markov random field,

$$\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^{-1}),$$

with sparse precision matrix

$$\mathbf{Q} = \frac{1}{1 - \rho^2} \mathbf{S} [(1 - \rho^2) \mathbf{I} + \rho^2 \mathbf{D} - \rho \mathbf{A}] \mathbf{S}. \tag{2}$$

Here $\mathbf{A}$ records unique firm–executive links, $\mathbf{D}$ records the corresponding node degrees, and $\mathbf{S} = \text{diag}(\sigma_a^{-1}\mathbf{I}_{N_f}, \sigma_z^{-1}\mathbf{I}_{N_m})$ is the inverse scale matrix. The precision matrix is sparse: its off-diagonal entries are nonzero only for linked firms and executives. This sparsity gives the Markov property. Conditional on directly connected neighbours, effects at more distant nodes are independent.

The model has four parameters, $\boldsymbol{\theta} = (\sigma_a, \sigma_z, \rho, \sigma_\varepsilon)$. On simple networks without cycles (“forests”), firm quality variance equals σ_a^2 , executive quality variance σ_z^2 , correlation ρ for every linked firm–executive pair, and the match-noise variance is σ_ε^2 . Cycles make the corresponding second moments depend on the network structure, but the numerical differences on our estimation graph are small. At the headline estimates, σ_a^2 and σ_z^2 differ from the exact graph-implied variance components by about 0.7% and 1.1%, and ρ differs from the implied correlation by 0.005. We therefore refer to the parameters as variances and correlation in the main text. The maximum likelihood estimator and the variance decompositions use the exact graph-implied moments.

The mobility graph and identification from paths

Figure 1 represents the data as a bipartite graph. Firms and executives are nodes; unique firm–executive links are edges. A path is a sequence of alternating firms and executives, and a connected component contains all nodes joined by such paths.

The identification argument is clearest on a forest, where any two connected nodes have a unique path. The correlation between latent effects at graph distance d is then ρ^d . For two spells whose executives are separated by h edges and whose firms lie on the connecting path,

$$\text{Cov}(y_{i_1 m_1}, y_{i_2 m_2}) = \rho^{h-2} (\sigma_a + \rho \sigma_z)^2, \quad (3)$$

so two additional hops multiply the covariance by ρ^2 , independently of the variance parameters.

Equation (3) separates the covariance level, which depends on σ_a , σ_z , and ρ , from the decay rate, which depends only on $|\rho|$. (The sign of ρ can be recovered jointly with the variance parameters.) Figure 2 evaluates this implication using centered covariance moments from balanced sets of outcome pairs on isolated shortest paths. We report separate series for paths with firm, executive, and mixed endpoints because their covariance levels differ. In all three series covariance declines with distance, but more slowly than the headline maximum-likelihood estimate implies. Outcomes therefore remain more strongly correlated over longer network distances than the fitted model

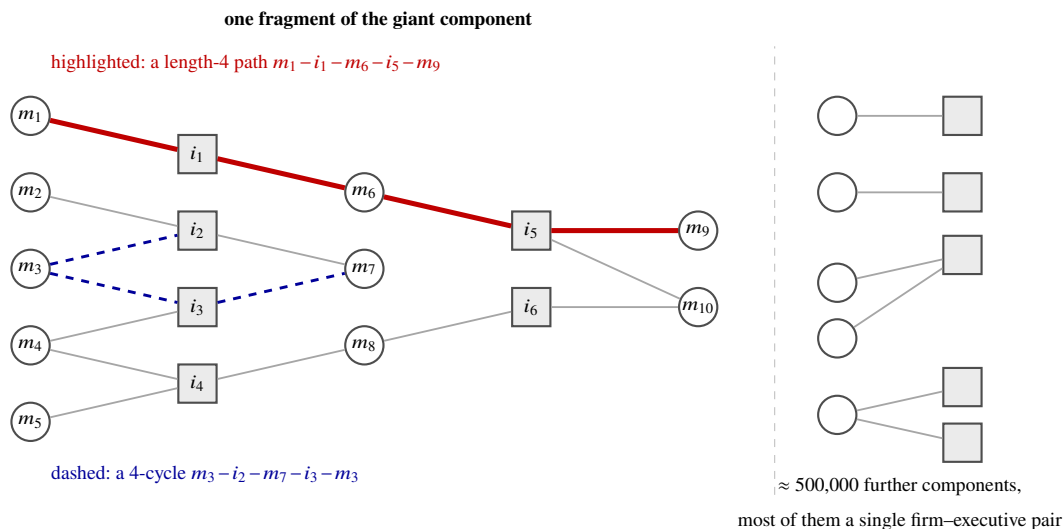


Figure 1: The bipartite executive–firm mobility graph. Circles are executives, squares are firms, and edges are unique firm–executive links. Left: a schematic fragment of the giant component, with one length-4 path highlighted in red and one 4-cycle dashed in blue. Right: the small components that make up the bulk of the graph. In the Hungarian data the graph has 514,085 connected components; the largest holds 42% of edges and the remainder are mostly isolated firm–executive pairs. **[NEED: this panel is a schematic. A readable subgraph extracted from the actual estimation graph would be better — src/plot/network.figure.jl currently produces degree/component/path-length histograms, not a subgraph rendering]**

predicts. This is a model-fit diagnostic: the maximum-likelihood estimator fits all outcomes on the full graph rather than targeting these selected path moments.

Data and the executive mobility network

Hungary provides an ideal setting for measuring executive sorting because administrative records cover firms beyond the listed sector. We combine the corporate-court registry of executives authorized to represent a company with annual balance-sheet filings for all firms with double-entry bookkeeping (19–21). The panel spans 1990 to 2022 **[NEED: the outline says 1992–2022; src/create/balance.sql filters year BETWEEN 1990 AND 2022 on the 2024 vintage of the balance data. Confirm the reported window]**. Materials and Methods describes entity resolution and sample construction.

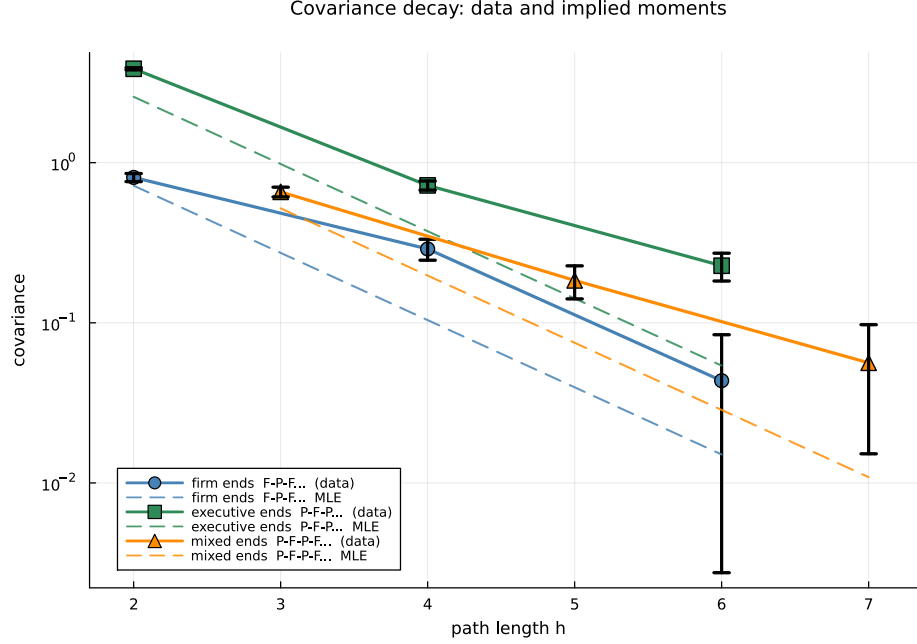


Figure 2: Empirical covariance decays more slowly than the fitted network model implies.

Markers show centered covariance between outcomes whose endpoint edges lie on isolated shortest paths of length h ; bars are 95% confidence intervals. Colors and marker shapes distinguish firm, executive, and mixed endpoint orientations. Dashed lines show the moments implied by the headline maximum-likelihood estimates. Dotted lines show a minimum-distance fit computed only on the selected path moments as a diagnostic, not as an alternative headline estimator. The vertical axis is logarithmic. Supplementary Material 1 describes the construction and derives the orientation-specific covariance formulas.

The registry records every officer authorized to represent a company, not only executives of listed firms. We identify chief executives from titles where available and from continuity and officer counts otherwise. The resulting population covers **[NEED: share of firms, employment, revenue in the economy]**, extending the analysis beyond listed companies, which account for only a small fraction of the Hungarian corporate sector.

Table 1 reports the network descriptives. Three features shape identification.

First, the graph is sparse and fragmented. It contains 514,085 connected components; the largest contains 386,015 executives and 341,650 firms joined by 823,158 spells. It holds 29.6% of executives and 33.0% of firms.

Second, the graph is close to a forest. Bridges account for 81.8% of spells: removing one disconnects its component. The remaining cycles are few and long.

Third, paths are long even inside the giant component. The shortest path between two randomly drawn executives averages 19.5 edges. A fixed-effects comparison of two typical firms therefore rests on roughly twenty links, allowing estimation error to accumulate along the chain.

Table 1: Descriptive statistics of the firm–executive mobility network.

	Value
<i>Full network</i>	
Executives	1,304,580
Firms	1,036,231
Firm–executive edges	1,937,552
Connected components	514,085
Bridges	1,584,810
Bridge share of edges	81.8%
<i>Giant component</i>	
Executives	386,015
Firms	341,650
Edges	823,158
Share of executives	29.6%
Share of firms	33.0%
<i>Giant component paths</i>	
Mean shortest path, executive pairs	19.5

Estimating on the full economy

We estimate on the full graph because restricting the sample to its largest connected component would select firms and executives with greater mobility. The likelihood factorizes by component, so each component contributes the moments it contains.

A one-firm–one-executive component contributes the moment $\mathbb{E}[y^2] = \sigma_a^2 + \sigma_z^2 + 2\rho\sigma_a\sigma_z + \sigma_\varepsilon^2$. It cannot identify ρ by itself, but it informs the four parameters jointly through outcome variance.

Table 2: Composition of the giant component versus all remaining components. Employment is a firm’s average headcount over the years it is observed; revenue is in logs. Manufacturing is TEAOR08 section C. Founder shares are over manager-firm matches.

	Giant	Other	All
<i>Counts</i>			
Firms	341,650	694,581	1,036,231
Executives	386,015	918,565	1,304,580
Match observations	1,116,741	1,409,277	2,526,018
<i>Shares of total (%)</i>			
Firms	33.0	67.0	100.0
Executives	29.6	70.4	100.0
Employment	55.9	44.1	100.0
<i>Firm characteristics</i>			
Employment, mean	9.1	3.5	5.4
Employment, median	1.4	1.2	1.2
Log revenue, mean	9.37	8.65	8.89
Manufacturing	8.5%	9.5%	9.2%
Founder match observations	46.8%	72.1%	60.9%

What it lacks is path information: without a second link, it provides no covariance at a longer network distance.

Supplementary Material 5 describes the conventional giant-component comparison.

Adding roughly 426,000 small components doubles the node count and raises the number of edges by 59%, but changes $\hat{\rho}$ by at most 0.011 across estimators and graph stages. We therefore use the full graph as our baseline.

Table 2 shows that the giant contains 33.0% of firms and 29.6% of executives, but 55.9% of employment. The difference is concentrated in the right tail: mean employment is 9.1 in the giant and 3.5 elsewhere, while the corresponding medians are 1.4 and 1.2. Revenue is similarly higher in the giant, but manufacturing shares differ by only one percentage point. Thus, the giant contains more large firms but is not a distinct sectoral stratum.

The largest difference concerns executive mobility. Founder matches account for 72.1% of links outside the giant, compared with 46.8% inside it. The small-component mass therefore contains many firms paired with executives who have not moved between firms. Component size is associated with firm size at the bottom of the distribution, but the association flattens well before the giant: mean employment rises from 2.76 in singleton firm–executive pairs to 5.87 in components with 11–50 nodes, then is 5.84 in larger non-giant components. Mean log revenue follows the same pattern, rising from 8.33 to 9.22 and then remaining 9.23. We do not assess regional composition because the administrative inputs contain no geography variable.

Sorting between executives and firms

Our preferred specification uses the full graph after the edge pruning described in Materials and Methods. The estimates are

$$\hat{\rho} = 0.322, \quad \hat{\sigma}_a = 1.388, \quad \hat{\sigma}_z = 0.394, \quad \hat{\sigma}_e = 1.348,$$

using $K = 1,693,263$ match observations between $N_f = 1,036,231$ firms and $N_m = 1,304,580$ executives, or 2,340,811 nodes in total. The sorting estimate is interior: $\hat{\rho} = 0.322$ lies below the graph’s feasibility ceiling, $\rho_{\max} = 0.401$. The corresponding marginal correlation between matched firm and executive effects is 0.327.

Our main uncertainty concerns the graph and observation specification rather than one fixed-sample calculation. We therefore emphasize the same-sample estimator comparison in Table 4 and the pruning path in Figure 3.

[NEED: Re-run all estimates on the 2024 balance vintage. Only the headline `nbedge-lnb3.0` fit is on the current panel; the pruning sweep, spanning forest, subperiod, and leave-out comparisons are on the earlier vintage. The vintage gap moved $\hat{\rho}$ from 0.280 to 0.327 on the identical specification.]

Variance decomposition

Table 3 decomposes revenue variance into four parts. Firm effects account for 47.7%, consistent with persistent productivity differences across firms (22). Executive effects alone account for 3.8%, and their covariance with firm effects adds 8.9%. The sorting covariance is more than twice the

Table 3: Variance decomposition of log real revenue. Components are the graph-implied unconditional variance contributions at the headline estimates.

Component	Variance (log revenue squared)	Share (%)
Firm effects	1.940	47.7%
Executive effects	0.157	3.8%
Sorting covariance	0.360	8.9%
Match noise	1.614	39.6%
Total	4.071	100.0%

executive variance, so executive identity matters more through the firms with which executives are matched than through executive dispersion alone. Match-specific noise accounts for the remaining 39.6%. The executive component is smaller than in the fixed-effects decomposition: on the common subgraph in Table 4, Panel B, executive variance falls from 4.4 under fixed effects to 0.023 under the GMRF.

On a forest, σ_a^2 and σ_z^2 are exactly the firm and executive variance components. Table 3 instead reports the exact unconditional components implied by the cyclic estimation graph. The numerical differences are small, but the table uses $\mathbf{Q}^{-1}$ so that the four entries add to total outcome variance exactly.

The economic importance of sorting depends on the graph-implied covariance, not on ρ alone. A market can sort strongly but gain little from sorting when executive heterogeneity is small. Random assignment removes this covariance and lowers aggregate revenue by 16.5%.

Robustness to network structure

The precision matrix is positive definite only when $|\rho|\lambda_{NB} < 1$, where λ_{NB} is the non-backtracking spectral radius (Materials and Methods). The raw Hungarian graph has $\lambda_{NB} = 15.26$, which restricts ρ to less than 0.066. This restriction comes from one dense core and survives conventional sample filters. We therefore remove individual edges from densely interconnected neighbourhoods until λ_{NB} falls below a target. Reaching $\lambda_{NB} < 3$ removes 34,500 of 1,732,154 edges, or 2.0%; it removes no nodes and leaves both the component count and the giant component’s share unchanged.

Figure 3 plots $\hat{\rho}$ against the resulting feasibility ceiling, $\rho_{\max} = 1/\lambda_{NB}$. The bound determines the estimate when the ceiling is low, but the estimate becomes interior and stable as the ceiling

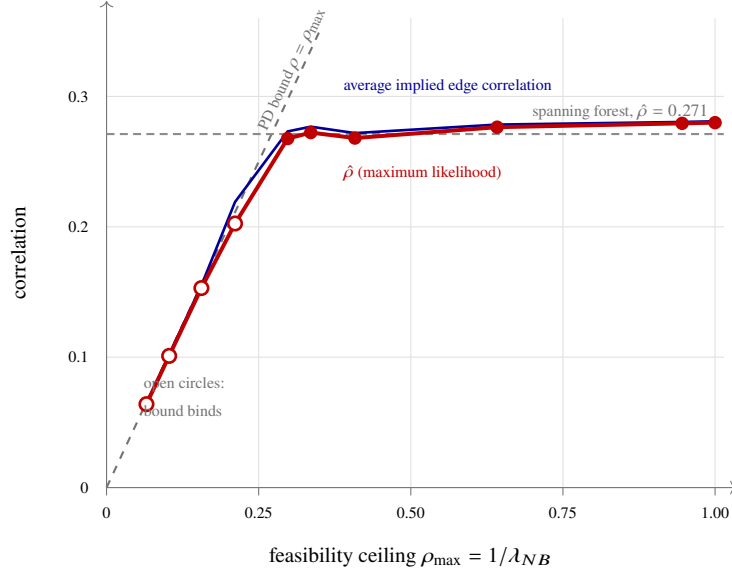


Figure 3: The estimate against the pruning-implied bound. Each point is one edge-pruned version of the same graph, indexed by the feasibility ceiling $\rho_{\max} = 1/\lambda_{NB}$ it achieves; the raw graph is the leftmost point ($\lambda_{NB} = 15.26$). Open circles mark fits censored at the bound. Once enough edges are pruned the bound stops binding and $\hat{\rho}$ flattens at an interior optimum; the spanning forest, which requires no pruning, lands on the same value. [NEED: drawn from output/fig/rho_vs_rhomax.csv, which is the mincomp-2-2 sample on the pre-refresh balance panel, and plateaus at $\hat{\rho} = 0.272$, not at the 0.33 headline. The pruning sweep has to be re-run on full/full with the refreshed panel — only the single nedge-lnb3_0 point exists there today.]

risers. The implied edge correlation follows the same plateau, as does the spanning-forest estimate. Hence the stabilized sorting estimate is not created by a particular pruning threshold.

Economic importance

We ask how much aggregate revenue would fall if the same executives were assigned randomly across the same firms. We hold the realized marginal distributions of firm and executive effects fixed and randomly reassign executives across firms, so the covariance between their effects is zero. This differs from changing ρ , which changes the model's entire covariance structure. Reassignment changes only the allocation of a fixed population of firm and executive effects.

The observed outcome is log revenue. Under a log-normal approximation, mean revenue is

$\mathbb{E}[R] = \exp(\mu + V/2)$. Removing the sorting contribution of 0.360402 lowers outcome variance by that amount and mean revenue by $0.360402/2 = 0.1802$ log units, or 18.0 log points: $1 - \exp(-0.1802) = 16.5\%$. A constant value-added share would not change the log-point effect. This exercise does not define a perfect-sorting counterfactual, because entangled network neighbourhoods make that allocation ill-defined; executive reassignment may also change the network itself.

Comparison with fixed-effects estimators

Table 4 first validates the GMRF in the dense Veneto Employer History, where worker mobility provides a useful fixed-effects benchmark. The naive plug-in correlation is -0.086 ; the homoscedastic correction raises it to 0.105, and leave-one-out raises it to 0.155. On the same 56,044 spells, the degree-weighted GMRF gives 0.160. A parametric bootstrap confirms this agreement in expectation but not in precision: the across-draw standard deviation of the implied correlation is 0.023 for the GMRF and 0.137 for leave-one-out.

Panel B repeats the comparison on the sparse Hungarian executive network. On the common subgraph, the naive fixed-effects estimate is -0.594 , leave-one-out raises it to -0.162 , but the GMRF gives $+0.351$. All three estimates use the same 91,085 matches; sample composition cannot explain the sign reversal.

Leave-one-out estimation nevertheless requires a highly selected graph. The match-level Leave-Out.jl procedure drops bridges in a single pass and then retains the largest 2-edge-connected block: 175,661 matched pairs, or 10.1% of pairs, involving 55,394 executives (4.6% of executives). The 91,085-match common graph in Table 4 is a further restriction for the same-sample comparison. The published KSS procedure is stricter: its iterated articulation-point pruning retains 59 person-years across six executives on the full network. These are different procedures, not competing estimates. Supplementary Material 6 reports the sample funnel and leverage diagnostics; (18) studies the mechanism behind the sparse-network divergence.

Table 4: Estimator comparison on dense and sparse mobility networks. Each column reports an estimator fit on the stated comparison sample.

Statistic	naive FE	homosced.	leave-out	weighted GMRF
<i>Panel A. Dense worker–firm replication data</i>				
Firm–worker corr.	−0.086	0.105	0.155	0.160
Job spells	56,044	56,044	56,044	56,044
<i>Panel B. Hungarian executive data, common subgraph</i>				
Firm variance	8.216	?	4.223	3.681
Executive variance	4.408	?	0.435	0.023
Firm–executive corr.	−0.594	?	−0.162	+0.351
Match-noise variance	—	?	1.495	1.656
Match observations	91,085	?	91,085	91,085

Notes: Panel A uses the leave-one-worker-out connected set from the worker–firm replication data. Panel B uses temp/samples/full/full/common-lnb2_5. Fixed-effects columns report moments of realized effects; the weighted GMRF column reports the unconditional decomposition implied by $\mathbf{Q}^{-1}$ on the same graph. The GMRF parameter ρ is shown as the correlation because its graph-implied marginal counterpart is numerically close on this sparse graph. [NEED: Re-estimate Panel B on the refreshed balance-panel vintage.]

Discussion

We find positive but incomplete assortative matching between executives and firms. The estimate, $\rho = 0.32$, is consistent with the sorting premise of the executive-pay literature, but well below the near-perfect sorting imposed by some assignment models. Importantly, we recover this correlation even though three quarters of executives never move to a second firm.

Sorting accounts for 8.9% of revenue variance. This contribution depends jointly on the correlation and executive heterogeneity; ρ alone is not a measure of misallocation. The subperiod estimates suggest that the covariance changes mainly through σ_z , making the covariance the relevant quantity for the random-assignment counterfactual.

The positive estimate reflects the method, not a different sample. On dense worker data the GMRF reproduces leave-one-out; on the sparse executive network the two estimators diverge even on identical observations. This pattern points to estimator-specific bias under sparse mobility, whose

mechanism is analyzed in (18). **[NEED: Add the event-study reconciliation once the companion $\text{Var}(\Delta Z)$ moment is available: $\text{Var}(Z_{\text{new}} - Z_{\text{old}}) = 2\sigma_z^2(1 - \rho_{ZZ})$, citing (23).]**

Firm heterogeneity remains the dominant source of revenue dispersion, accounting for 47.7%. Executive dispersion accounts for 3.8%, while the 8.9% sorting covariance shows that executives matter primarily through their allocation across firms. The GMRF attributes less executive variation to persistent individual effects than the naive fixed-effects decomposition because it accounts for network dependence and limited mobility (7, 10, 24–28).

Match-specific noise accounts for 39.6% of revenue variance. This component bundles idiosyncratic shocks, tenure effects, within-match learning, and skill acquisition. We leave their separation for future work.

We view the GMRF as a complement to equilibrium matching models, not a replacement for them. Such models derive sorting from complementarities, preferences, and search frictions (29–32); our reduced-form model measures sorting from production outcomes on the realized bipartite graph.

Supplementary Material 7 provides a network-formation interpretation. Under independent population types and a sparse-link approximation, the estimated mean and covariance parameters determine relative link propensities after normalizing the complementarity in match surplus. The positive-link likelihood does not model the complete risk set, but it offers a route from network measurement to a structural model of executive allocation.

Materials and Methods

Statistical implementation

The main text defines the outcome model, Gaussian field, and precision matrix. We now describe likelihood evaluation, precision weighting, graph restrictions, and construction of the administrative panel.

From latent effects to observed outcomes

We link latent effects $\mathbf{x}$ to observed outcomes $\mathbf{y}$ by writing Equation (1) in matrix form:

$$\mathbf{y} = \mathbf{V}\mathbf{x} + \boldsymbol{\varepsilon}. \quad (4)$$

The $K \times (N_f + N_m)$ design matrix $\mathbf{V}$ maps match observations to latent types. A single-executive row loads once on its firm and once on its executive. A cluster with q simultaneous executives loads once on the firm and $1/q$ on each executive, so the cluster contributes one outcome and one error term. With $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I}_K)$, the implied outcome covariance is

$$\boldsymbol{\Omega} = \mathbf{V}\mathbf{Q}^{-1}\mathbf{V}^\top + \sigma_\varepsilon^2 \mathbf{I}_K. \quad (5)$$

This $K \times K$ matrix is dense, making direct inversion infeasible.

Maximum likelihood and computation

The log-likelihood is

$$\ell(\boldsymbol{\theta}) = -\frac{K}{2} \ln(2\pi) - \frac{1}{2} \ln \det \boldsymbol{\Omega} - \frac{1}{2} \mathbf{y}^\top \boldsymbol{\Omega}^{-1} \mathbf{y}. \quad (6)$$

Here $\boldsymbol{\theta} = (\sigma_a, \sigma_z, \sigma_\varepsilon, \rho)$. The Woodbury identity replaces operations on $\boldsymbol{\Omega}$ with operations on $\mathbf{P} = \mathbf{Q} + \sigma_\varepsilon^{-2} \mathbf{V}^\top \mathbf{V}$, an $(N_f + N_m) \times (N_f + N_m)$ matrix. For single-executive observations, $\mathbf{V}^\top \mathbf{V}$ adds diagonal counts and firm–executive entries. Grouped teams also add small executive–executive blocks. Thus $\mathbf{P}$ has the union of prior-graph and within-team sparsity and remains sparse even though $\boldsymbol{\Omega}$ is dense. Sparse Cholesky factorizations then supply the log-determinant and quadratic form. The headline fit contains 2,340,811 nodes and converges in 128 iterations and 518 seconds. We enforce positive definiteness through the constraint below, without a ridge. When Cholesky fill-in is costly, we approximate the log-determinant with the Hutchinson trace estimator (33) and stochastic Lanczos quadrature. These methods require only sparse matrix–vector products and scale linearly with the number of edges. We construct graphs with (34).

[NEED: document optimizer settings, starting values and convergence tolerances properly, and state per table which results use the exact and which the stochastic path.]

Precision weighting and graph restrictions

Equation (2) adjusts the diagonal for node degree. Under the plain precision $\mathbf{S}(\mathbf{D} - \rho\mathbf{A})\mathbf{S}$, marginal correlations vary with degree. Under the variance-stable precision, every linked pair on a forest has correlation ρ , while firms and executives have marginal variances σ_a^2 and σ_z^2 . Across 20,000 sampled forest edges, the numerical discrepancy from this identity is below 7×10^{-15} .

The observed graph departs from a forest in four ways. The likelihood accounts for each departure; the forest formulas provide intuition, not an approximation used in estimation.

Cycles. Alternative paths raise covariance, so the average edge correlation exceeds ρ . In the data, 1,463,168 of 1,730,350 matches, or 85%, are bridges; the remaining cycles are few and long. On the estimation graph, a tree edge has implied correlation $\hat{\rho}$ to four decimals, while the median edge in the cyclic core lies only 0.019 above it. Across all edges, the implied correlation is 0.277 when $\hat{\rho} = 0.272$ on the corresponding sample.

Positive definiteness. Equation (2) is positive definite only when $|\rho|\lambda_{NB} < 1$, where λ_{NB} is the non-backtracking spectral radius. The mathematical ceiling is $\rho_{PD} = 1/\lambda_{NB}$; the optimizer stays inside it using the conservative box $\rho_{\text{box}} = \min\{0.95, 0.98/\lambda_{NB}\}$. We compute λ_{NB} from the Ihara–Bass companion matrix rather than construct the larger directed-edge matrix. Because the bound depends only on the graph, not on outcomes, we prune individual edges rather than delete entire motifs. Few non-backtracking eigenvalues exceed one, and the leading eigenvector is concentrated on roughly a dozen nodes; a small region therefore drives the constraint. Figure 3 reports sensitivity to the pruning level.

Missing outcomes. We retain spells without observed firm performance in the graph because they connect neighbouring firms and executives even though they contribute no outcome. Such spells affect degrees and the prior, but not the likelihood. In the headline graph, 198,183 edges are graph-only; they account for 291,386 of 1,984,649 spell rows (14.7%). The match pipeline retains these edges through sampling and spectral pruning, so the precision matrix is constructed from the same graph used to calculate the feasibility bound. Dropping them would distort the path structure that identifies ρ : in the 2015–2017 window, it removes 134,442 nodes and more than halves the giant component.

Multiple simultaneous executives. A match cluster is a firm interval over which the set of co-

present executives is constant. When a cluster has several executives, its design row loads once on the firm effect and on the average of the executives' effects. It therefore has one error term, rather than crediting every executive with the full firm outcome. Supplementary Material 3 derives this grouped observation; Supplementary Material 4 explains its effect on $\mathbf{V}^\top \mathbf{V}$.

Observables

Observable characteristics enter as

$$y_{im} = \mathbf{w}_{im}^\top \boldsymbol{\beta} + a_i + z_m + \varepsilon_{im},$$

where $\mathbf{w}_{im}$ contains industry and year effects. Observables shift the mean without changing the covariance structure. We can therefore residualize outcomes in a first step or profile $\boldsymbol{\beta}$ from the Gaussian likelihood. The reported estimates demean log revenue by year and two-digit industry.

Data sources and construction

We combine two administrative sources covering Hungarian firms (19, 20). The corporate-court registry records all executives authorized to represent a company, including chief executives and other officers with signatory authority (21). Records are spells that end when a position or identifier changes. They contain names, addresses, mother's names from 1999, birthdates from 2010, and numerical identifiers from 2013. Annual balance sheets for all firms with double-entry bookkeeping provide revenue, employment, sector, and ownership. Together, the sources cover more than one million firms.

We link executives across firms and time by entity resolution. Numerical identifiers provide direct links from 2013 onward. In earlier years, we construct person identifiers from names, addresses, mother's names, and birthdates. Link quality improves after mother's names enter in 1999 and birthdates in 2010; names and addresses provide the links before then.

We identify chief executives among registered representatives in three steps. First, we use managing-director titles where available. Second, we classify all representatives as chief executives when a firm has no more than three. Third, for firms with more representatives, we use continuity with previously identified chief executives.

We apply five sample restrictions. First, we exclude mining and finance because they use specialized accounting. Second, we drop firms that ever have more than two simultaneous chief executives. Third, we remove firms with more than ten executive changes, which are likely to contain classification error. Fourth, we exclude state-owned enterprises. Fifth, we exclude public firms, joint-stock companies, and cooperatives because their governance differs from that of private corporations. We retain private nonemployer firms because single-person companies form a substantial part of the Hungarian corporate sector.

The outcome is log real firm revenue, deflated by standard price indices and demeaned by year and two-digit industry. We first average revenue over the relevant firm interval. Intervals with the same firm and executive set form one match cluster and contribute one grouped outcome to the likelihood.

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Data, code, and materials availability: The restricted-use Cégjegyzék LTS and Mérleg LTS source data are distributed by HUN-REN KRTK and published by Opten Zrt. (19, 20). Opten has authorized release of the study’s full pseudonymized edge list, containing pseudonymous firm identifiers, pseudonymous executive identifiers, and incrementing match identifiers. Before publication, we will deposit that edge list, all data-cleaning code, all analysis code, and documentation in a public DOI-minting repository such as Zenodo or Figshare. No physical materials were generated in this work.

Supplementary materials

Materials and Methods

Supplementary Text

Figs. S1 to S[**NEED: final figure number**]

Tables S1 to S[**NEED: final table number**]

References (*[NEED: supplement-only reference range]*)